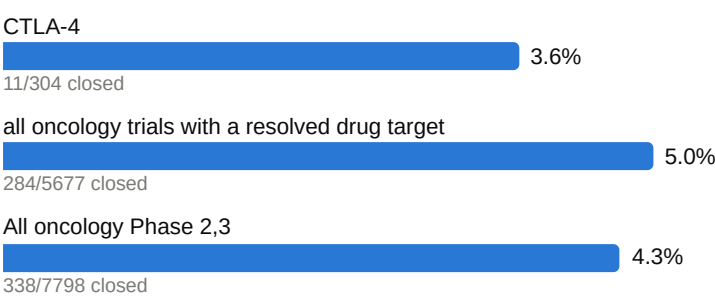


CTLA-4: 3 molecules stopped early, all monoclonal antibodies

11 stopped trials, but 3 molecules — all monoclonal antibodies: Ipilimumab, Erfonrilimab and Tremelimumab. Against 5.0% for all oncology trials with a resolved drug target and 4.3% across all oncology Phase 2,3 trials in the same window. Rates count trials that stopped early for efficacy, safety or benefit–risk reasons; trials that completed and missed their endpoints are not counted.

3.6% 11 of 304 closed trials (95% CI 2.0%–6.4%)	10 / 1 / 0 efficacy / safety / both (adds to 11)	40.0% of 760 trials have closed · 456 still open or unresolved	1.1% stopped for a biological reason within 3 years of starting (95% CI 0.5%–2.4%; 404 still at risk; 1.5% area-wide)
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How this compares



What is in this class: 286 distinct experimental drugs across 760 trials (antibody 754, small molecule 252, unknown 91, adc 32). Most tested: IPILIMUMAB (349), NIVOLUMAB (295), CADONILIMAB (139), DURVALUMAB (129), TREMELIMUMAB (129). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 11 stops came from 11 sponsor-asset programmes across 9 sponsors; the largest (Weill Medical College of Cornell University) contributed 1; removing that programme's trials from both sides leaves 3.3%. Screened alongside 83 other segments in this area, it does not survive a 10% false-discovery correction (q=1, Benjamini-Yekutieli). A further 8 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 6.2%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	0	142	0.0%	0.0%–2.6%
2018–2020	8	111	7.2%	3.7%–13.6%
2021–2024	3	51	5.9%	2.0%–15.9%

Where the stops sit

Efficacy / futility: 10, Safety: 1.
Sponsors with most stops: Herlev Hospital (2), Jiangsu Alphamab Biopharmaceuticals Co., Ltd (2), M.D. Anderson Cancer Center (1), Weill Medical College of Cornell University (1), University of Washington (1), H. Lee Moffitt Cancer Center and Research Institute (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT03274258	2	M.D. Anderson Cancer Center	IPILIMUMAB, NIVOLUMAB	2025-11	Efficacy	The study met its prespecified futility criteria during the prespecified interim futility analysis after enrolling the first cohort of 10 patients.
NCT05116917	2	Herlev Hospital	NIVOLUMAB, IPILIMUMAB, Influenza vaccine	2023-10	Efficacy	Based on the prespecified interim analysis the predictive probability was below 10% for the prespecified endpoint.
NCT04925947	2	Weill Medical College of Cornell University	ERFONRILIMAB	2023-05	Efficacy	Data collected did not support the study endpoints
NCT05001724	3	Jiangsu Alphamab Biopharmaceuticals Co., Ltd	ERFONRILIMAB, LENVATINIB	2023-02	Efficacy	Because the efficacy data did not meet expectations, the sponsor decided to terminate the study.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT03838848	2	Jiangsu Alphamab Biopharmaceuticals Co., Ltd	ERFONRILIMAB, CARBOPLATIN, PEMETREXED, NINGETINIB	2022-07	Efficacy	Cohort A,B,C end enrollments. Cohort D and E were considered to have no significant clinical benefit at the SMC meeting and decided to terminate enrollment.
NCT03522584	2	University of Washington	DURVALUMAB, TREMELIMUMAB	2021-12	Efficacy	Terminated by the sponsor due to lack of efficacy in study treatment.

The 6 most recent of 11 stopped trials. Every molecule behind all 11 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the evidence package.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated).
Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: trials that completed with negative results are not counted, and programs discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 40.0% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

Any mechanism, sponsor or indication, updated weekly. The dataset behind this brief covers every stopped oncology trial with an efficacy or safety signal plus the full denominator universe. Free sample and licensing: clinicaltrialfailures.com/data-licensing

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.